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Abduction and Deduction in Dynamical Cognitive Science

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Abstract

This paper reviews the recent history of a subset of research in dynamical cognitive science, in particular that subset that allies itself with the sciences of complexity and casts cognitive systems as interaction dominant, noncomputational, and nonmodular. I look at this history in the light of C.S. Peirce’s understanding of scientific reasoning as progressing from abduction to deduction to induction. In particular, I examine the development of a controversy concerning the use of the interaction dominance of human cognitive systems as an explanation of the ubiquitous $1/f$ noise, multifractality, and complexity matching in human behavior.

Keywords: Complex systems; Dynamical systems; Philosophy of science

1. Introduction

One of the ways in which van Gelder’s *Behavioral and Brain Sciences* paper from (1998) is an advance over his paper from a few years earlier in the *Journal of Philosophy* (1995) is that it uses the Haken–Kelso–Bunz model (HKB, 1985) as its primary example. The earlier

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paper used Busmeyer and Townsend's (1993) decision field theory—essentially a stochastic neural network, feeding into a set of modules of the variety familiar from classical cognitive science. By using HKB, van Gelder showed that dynamical cognitive science is a big tent and that in many cases dynamical systems models in cognitive science imply a different kind of explanation than the computational explanations of traditional cognitive science, allying the dynamical cognitive sciences with complexity theory. At the same time, however, the complex systems version of dynamical cognitive science has proven controversial. In this paper, I trace some of this controversy. To begin, I describe American philosopher C.S. Peirce's views on abduction in the sciences. I then show how abductive reasoning is used by both sides of the controversy over complex systems explanations in the cognitive sciences. Finally, I point to some methodological advances and empirical evidence that bear on the controversy. Ultimately, I will argue that the complex systems variety of dynamical cognitive science is alive and well in the 2020s and that this does not imply that other approaches are moribund.

2. (Frankfurt on) Peirce on abduction

Charles Saunders Peirce's work on abductive inferences in the sciences was brought into the cognitive sciences by Jerry Fodor (1983; 2000). Fodor argued that *abduction* shows that cognition cannot be (only) computation. Though widely disputed, this is a rare point of agreement between Fodor and those outside the mainstream of cognitive science like phenomenological philosopher John Haugeland (1998) and ecological psychologist Michael Turvey (2018). Cognition, these disparate thinkers agree, can only be feasibly understood as computation if it involves informational encapsulation, that is, modules. Abduction seems to defy informational encapsulation and involve creative synthesis. What, then, is abduction? Peirce's own writing on this subject notoriously convoluted and jargon laden, so I will draw on a classical presentation of Peirce's work on scientific reasoning by Harry Frankfurt (1958). According to Peirce, according to Frankfurt, there are three basic kinds of scientific inference: abduction, deduction, and induction. In abduction, one hypothesizes an explanation for a surprising observation. It has the following form:

1. The surprising fact, F, is observed.
2. But if explanation E were true, F would be a matter of course.
3. Hence, there is reason to suspect that E is true.

This kind of inference to the best explanation is a crucial part of both science and everyday cognition. While preparing to cook dinner, we notice that the tofu is missing from the refrigerator and try to come up with an explanation for that fact. It would be unsurprising for the tofu to be gone if some member of our family had made miso soup for lunch today. We reasonably come to suspect that someone made miso soup for lunch. In a more scientific context, we notice that dolphins, unlike most other sea creatures, frequently come to the surface and seem to breathe air. One possible explanation for this is that dolphins, despite their appearance and life in the ocean, are not fish. Abduction involves creativity and bringing to

bear arbitrary pieces of information from across subject areas. Abductive inference will be central to the disputes described below.

Peirce also identified two further, more familiar types of scientific and everyday inferences, which follow abduction in attempting to gather scientific evidence. First, there is *deduction*, which takes the following form:

1. Derive further testable consequences C of explanation E.
2. Test those consequences C.
3. Observing C is evidence for E.

Deduction will be familiar to those who have studied the philosophy of science—it is often called the “deductive-nomological model of explanation” (Hempel and Oppenheim, 1948). The third type of scientific reasoning Peirce identifies is *induction*. In induction, one iterates the deductive process and decides on the correctness of explanatory hypothesis E, based on some criterion for the required number of observations of C to make it a matter of course. So, for Peirce, the scientific process starts with abduction, in which we generate an explanation for some phenomenon, moves on to deduction, in which we attempt to generate evidence that the explanation is the correct one, and finally, to induction, where we attempt to decide whether the evidence we have gathered is sufficient to demonstrate that the explanation is correct. In the rest of this paper, I will show this pattern—abduction to deduction to induction—at play in dynamical cognitive science.

3. Complex systems and dynamical cognitive science

The HKB that van Gelder discussed uses the physical properties of coupled oscillators to explain self-organizing patterns in human bimanual coordination. In earlier experiments, Kelso (1984) showed that human participants asked to tap the index fingers both of their hands along with a metronome, invariably tapped them in one of two patterns. Either they tapped the fingers in phase, with the same muscle groups in each hand performing the same movement at the same time, or out of phase, with the same muscle groups in each hand “taking turns.” Kelso found that out-of-phase movements were only stable at slower metronome frequencies; at higher rates, participants could not maintain out-of-phase coordination in their finger movements and transferred into in-phase movements. Haken, Kelso, and Bunz modeled this behavior using the following equation, a potential function:

$$V(\phi) = -A \cos \phi - 2B \cos 2\phi,$$

where ϕ is the relative phase of the oscillating fingers and $V(\phi)$ is the energy required to maintain relative phase ϕ . Relative phase is a collective variable, capturing the relationship between the motions of the two fingers. In-phase tapping, with both fingers making the same movements at the same time, has relative phase of 0, while out-of-phase tapping, with fingers taking turns, is relative phase $\pm\pi$. The transition from relative phase π to relative phase 0 at higher metronome frequencies is modeled by the derivative of relative phase,

$$\dot{\phi} = -A \sin \phi - 2B \sin 2\phi.$$

In these equations, the ratio B/A is a control parameter, a parameter for which small quantitative changes to its value lead to large qualitative changes in the overall system. These equations describe all the behavior of the system. These equations had already been applied to a wide range of human behavior when van Gelder wrote his article; that has continued in the 21st century.

In a recent retrospective on coordination dynamics, Scott Kelso, the K of the HKB model, explains the connection between HKB and self-organization.

Not only did HKB promote the mechanism of self-organization as an explanation of how new patterns of biological coordination could arise, persist and change, it brought to the fore the crucial roles of stability and variability. Up to then, variability was often seen as just “noise,” something to be damped out or ignored. Now, experiments and theory were saying something else entirely, namely that variability is ubiquitous and essential for biological coordination. Among other aspects, variability allows the system to adapt and to change flexibly. (Kelso, 2021, 309)

Kelso’s point is that the HKB model is not just an exercise in curve fitting in which we take a theory-free approach to capturing regularities in data. It takes parts of living things, from neurons to limbs, to be self-organized collections of strongly coupled oscillators. Given what physicists have said about coupled oscillators, going all the way back to Huygens and his observation of odd sympathy in pendulum clocks in the 1600s, HKB makes a series of predictions about the coordinated entities to which it is applied. One of these predictions is that there will be *critical fluctuations* as the coordinated system approaches a critical point, for example, just before it transitions from out-of-phase to in-phase coordination patterns. This prediction of critical fluctuation, which has been verified repeatedly in the cognitive sciences, is perhaps the most clear way in which the use of dynamical models like HKB allies the cognitive sciences with the sciences of self-organization and complexity.

A self-organizing system is a system that exhibits regularities that arise without a plan or leader but emerge from the interactions of the parts of the system. The posit that human action is self-organizing began the recent resurgence of interest in complex-systems dynamical systems models in psychology, starting with work by Peter Kugler, Kelso, and Turvey (1980; Kelso, Holt, Kugler, & Turvey, 1980). Dynamical systems models work by assuming that thinking, experiencing, and acting humans are self-organizing dynamical systems that comprise portions of their brains, bodies, tools, and sometimes other people. Self-organizing systems have their organization without a plan or controller. They can do so by using energy from their surroundings to create patterns, as Huygens’ pendulum clocks do with gravity and HKB’s wagging fingers do from the body’s metabolism of sugars. Consider, to take another mundane example, the whirlpool pattern that forms when a toilet is flushed. The water molecules temporarily move in a coordinated way that is very atypical for water molecules. They can exhibit this pattern of behavior because potential energy from the water in the tank is released when the toilet is flushed. This energy enables the pattern to form and last until the energy is dissipated, after which the water molecules go back to their more typical behavior. Physicists who work on self-organization would call this behavior of water molecules in a

flushing toilet a *synergy* (Haken, 1973; Kelso, 2009). A synergy is a collection of components that act as a temporary self-organizing unit, and in which the components exhibit behavior that is constrained by being parts of the synergy.

Synergies seem odd from a mechanistic point of view: a synergy is an aggregate entity that constrains the behavior of the smaller-scale components that make it up. Synergies, thus, imply a kind of downward or circular causation that strains our ordinary understanding of causes and effects. Nonetheless, there is nothing strange or magical about synergies—they occur routinely in nature and are well understood by physicists. The pattern in your flushing toilet is a synergy; convection rolls in boiling oil are synergies; bird flocks are synergies. In each of these cases, components form leaderless, temporary coalitions in which the behavior of the components (molecules, birds) is constrained by being part of the whole (the convection, the whirlpool, the flock) that they make up.

One of the key ways to identify synergies is that they are temporary coalitions that exhibit *interaction-dominant dynamics*, and are, therefore, *interaction-dominant systems* (van Orden, Holden, & Turvey, 2003, 2005; Riley & Holden, 2012). This technical term can be read literally: A system exhibits interaction-dominant dynamics when the interactions among the components dominate or override the dynamics that the components would exhibit separately. Systems with interaction-dominant dynamics (i.e., interaction-dominant systems) are genuinely unified systems. Over the last few decades, it has been demonstrated that many well-functioning physiological systems are interaction-dominant systems. Examples of this include human heartbeats (Peng et al., 1995) and gait patterns (Hausdorff, Peng, Ladin, Wei, & Goldberger, 1995), indicating that the chambers of the heart and the locomotory system are interaction-dominant systems, which is to say they are synergies.

To repeat: A *synergy* or *interaction-dominant system* is a highly interconnected system, each of whose components alters the dynamics of many of the others to such an extent that the effects of the interactions are more powerful than the intrinsic dynamics of the components (van Orden, Holden, & Turvey, 2003). In an interaction-dominant system, inherent variability (i.e., fluctuations or noise) of any individual component A propagates through the system as a whole, altering the dynamics of the other components, say B, C, and D. Because of the dense connections among the components of the systems, the alterations of the dynamics of B, C, and D will lead to alterations to the dynamics of component A. That initial random fluctuation of component A, that is, will reverberate through the system for some time. So too, of course, would nonrandom changes in the dynamics of component A. This tendency for reverberations gives interaction-dominant systems what is referred to as “long memory” (Chen, Ding, & Kelso, 1997). In contrast, a system without this dense, cyclical feedback, a *component-dominant system*, would not show this long memory. For example, imagine a computer program that controls a robotic arm. Traditionally, this is called a *controller-plant* system. Although noise that creeps into the commands sent to from the controller (i.e., the computer) to the arm (i.e., the plant) might lead to a weld that misses its intended mark by a few millimeters, that missed weld will not alter the behavior of the program when it is time for the next weld. Component-dominant systems do not have long memory. Moreover, in interaction-dominant systems, one cannot treat the components of the system in isolation: because of the widespread feedback in interaction-dominant systems, one cannot isolate

components to determine exactly what their contribution is to a particular behavior. And because the effects of interactions are more powerful than the intrinsic dynamics of the components, the behavior of the components in an interaction-dominant system is not predictable from their behavior in isolation or from their behavior in some other interaction-dominant system. Interaction-dominant systems, in other words, are not *modular*. Indeed, they are in a deep way unified in that the responsibility for system behavior is distributed across all the components.

As will be discussed below, this unity of interaction-dominant systems has been used to make strong claims about the nature of human cognition (van Orden, Holden, & Turvey, 2003, 2005; Holden, van Orden, & Turvey, 2009; Riley & Holden, 2012). To use the unity of interaction-dominant systems to make strong claims about cognition, one must argue that cognitive systems are, at least sometimes, interaction dominant. This is where the critical fluctuations posited by the HKB model come in. Interaction-dominant systems self-organize, and some will exhibit *self-organized criticality*. Self-organized criticality is the tendency of an interaction-dominant system to maintain itself near critical boundaries, so that small changes in the system or its environment can lead to large changes in overall system behavior. It has long been known that self-organized critical systems exhibit a specific, fractal variety of fluctuation called *1/f noise* or *pink noise* (Bak, Tang, & Wiesenfeld, 1987). *1/f noise* or *1/f scaling* or *pink noise*—these terms are used synonymously in the literature—is a kind of not-quite-random, correlated noise, halfway between genuine randomness (white noise) and a drunkard's walk, in which each behavior is constrained by the prior one (brown noise). *1/f noise* is often described as a fractal structure in a time series, in which the variability at a short time scale is correlated with variability at a longer time scale. (This is the long memory of interaction-dominant systems described above.) In fact, interaction-dominant dynamics predicts that this *1/f noise* would be present. As discussed above, the fluctuations in an interaction-dominant system percolate through the system over time, leading to the kind of correlated structure to variability that is *1/f noise*. Here, as we will see, is the way to infer that cognitive and neural systems are synergies: exhibiting *1/f noise* is evidence that a system is interaction dominant; interaction-dominant systems are synergies. The mounting evidence that *1/f noise* is ubiquitous in human physiological systems, behavior, and neural activity is also evidence that human physiological, cognitive, and neural systems are interaction dominant, which, in turn, is evidence that they are synergies.

4. Abduction and deduction in dynamical cognitive science

In a 2003 paper, Guy van Orden, Jay Holden, and Turvey offer a version of Peirce's abductive inference based on the "surprising fact" of the ubiquity of *1/f noise* in human behavior.

1. Ubiquitous *1/f noise* in human behavior is observed.
2. But if the human cognitive system is interaction dominant, *1/f noise* in human behavior would be a matter of course.

3. Hence, there is reason to suspect that the human cognitive system is an interaction-dominant system.

Here, the proposed explanation for the ubiquity of $1/f$ noise is that the human cognitive system is interaction dominant. Van Orden et al. then engage in a Peircean deduction.

1. A consequence of the interaction dominance of the human cognitive system is that we should also see $1/f$ noise in cognition.
2. Test this consequence by looking for $1/f$ noise in human cognition.
3. Observing $1/f$ noise in human cognition is evidence that human cognitive systems are interaction dominant.

Van Orden et al. did two experiments to look for $1/f$ noise in human cognition. In one, participants were asked to say /ta/ in response to a visual cue for 45 practice trials and then 1100 test trials; in the second, participants were asked to say words that appeared on a monitor for 45 practice trials and 1100 experimental trials. The data in both cases was a time series of reaction times. Two quick points. First, it might seem strange that there were so many trials per participant; this was necessary because the time series analyses used to test for $1/f$ noise generally require comparisons of three orders of magnitude (Riley & Holden, 2012). Second, unlike the finger-wagging that was modeled by HKB, responding to a stimulus and reading words are straightforwardly cognitive activities and are tasks commonly used in experiments across the cognitive sciences. The findings in both van Orden et al.'s experiments were that participant behavior exhibited $1/f$ noise, providing evidence that human cognitive systems are interaction dominant.

That the human cognitive system is interaction dominant is a very strong claim. For one, it supports the claims discussed above by Fodor, Turvey, and Haugeland that cognition could not be (just) computation. For Fodor, the claim is that peripheral cognition might be modular and computational, but what he calls "central cognition" cannot be. For Turvey and Haugeland, the claim is that none of cognition is well understood as modular or computational.

5. An alternative abduction and deduction

One of the things we teach introductory psychology and cognitive science students is that an appropriate response to experimental evidence is to propose an alternative hypothesis that might explain the same data. This was exactly the response to van Orden et al. from Wagenmakers, Farrell, and Radcliff (2005). Van Orden et al. explained that $1/f$ noise in cognition appears because cognitive systems are interaction dominant. Wagenmakers et al. responded that we might see $1/f$ noise in a modular component-dynamic cognitive system because when the modules generate white noise, adding the white noise of each module together, the system might produce $1/f$ noise at its output. Here is their abduction:

1. Ubiquitous $1/f$ noise in human behavior is observed.

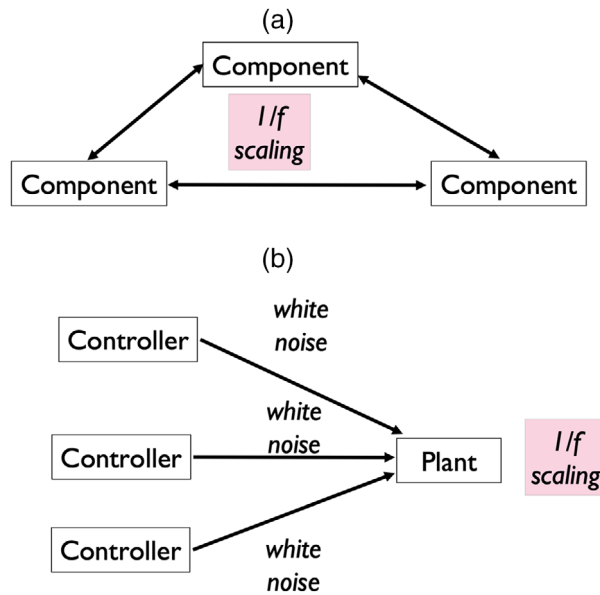


Fig. 1. Graphical representations of the alternative explanations of 1/f noise in human cognition. Fig. 1a (top) depicts van Orden et al.'s explanation of 1/f noise as the result of an interaction-dominant cognitive system. Fig. 1b depicts Wagenmakers et al.'s autorecurrent component-dominant alternative.

2. But if the human cognitive system is component dominant and each component contributes uncorrelated white noise to behavior, 1/f noise in human behavior would be a matter of course.
3. Hence, there is reason to suspect that the human cognitive system is a component-dominant system with each component contributing uncorrelated white noise to behavior.

(See Fig. 1.) As evidence for this hypothesis, the deductive step, Wagenmakers et al. show via simulations that a component-dominant system in which each component contributes white noise can generate data with 1/f noise.

Van Orden, Holden, and Turvey (2005) respond to Wagenmakers et al. by arguing that their simulations are gerrymandered and implausible. Their "newly proposed encapsulated mechanisms lack physical or psychological motivation before the fact and require that one posits implausible coincidences after the fact." This may indeed be the case, but it does not keep the Wagenmakers et al. abduction from presenting a live alternative hypothesis. And, importantly, an alternative hypothesis in the science is not the same as a refutation; it is an invitation to further research.

6. Further research

As an attempt to mediate this dispute, Ihlen and Vereijken (2010) re-examine both the van Orden et al. human data and the Wagenmakers et al. simulation data using a wavelet-based multifractal analysis. What they found was that although both the human and simulation data do exhibit $1/f$ noise, only the human data exhibits *multifractality*. The simulation data is monofractal, in that it exhibits the same degree of $1/f$ noise across all time scales. In contrast, the human data is multifractal in that it exhibits different degrees of $1/f$ noise at different time scales. The upshot of this work is that $1/f$ noise is not sufficient for claiming that a system is interaction dominant. Instead, the standard ought to be multifractality. This methodological innovation changes the requirements for arguing that the human cognitive system is interaction dominant, and so not modular or computational, from exhibiting $1/f$ noise to exhibiting multifractality.

In a related development, dynamical cognitive scientists have explored the relationship between multifractality and *complexity matching*. In complexity matching, two systems maximize the information transfer between them by coordinating the degree of their $1/f$ noise across multiple temporal scales (West, Geneston, & Grigolini, 2008). Complexity matching is a consequence of multifractality (Stephens & Dixon, 2011; Delignières, Almurad, Roume, & Marmelat, 2016); for two interacting cognitive systems to be able to match their degree of $1/f$ noise, they must be capable of organizing their behavior so as to control their degree of $1/f$ noise over time in concert with their partner. This in turn requires that cognitive systems be capable of moving from one degree of $1/f$ noise to another in their activity; that is, it requires that their performance is multifractal. So, complexity matching is evidence of multifractality, which makes it also evidence of interaction dominance.

This leads to a revision of the van Orden et al. abduction.

1. Multifractality is observed in human behavior.
2. But if the human cognitive system is interaction dominant, multifractality in human behavior would be a matter of course.
3. Hence, there is reason to suspect that the human cognitive system is an interaction-dominant system.

Rigoli, Holman, Spivey, and Kello (2014) use complexity matching to develop what might be called an *experimentum crucis*, explicitly designed for deciding between the Van Orden et al. and Wagenmakers et al. abductions. They divided participants into two groups and asked them to tap a keyboard spacebar along with a metronome, with one group tapping on beat and the other in syncopation for 1100 taps in an approximately 15-min trial. They measured time series of pupil dilations, heartbeat intervals, keypress durations, and keypress deviations. The van Orden et al. abduction and the Wagenmakers et al. abduction each imply different results. If human cognition is interaction dominant, we should expect each time series to separately exhibit $1/f$ noise and for the physiological measures (pupil dilation, heartbeat interval) and the behavioral measures (keypress duration and interval), respectively, to match one another's complexity. If human cognition is component dominant, we should expect neither $1/f$ noise in each measurement, nor complexity matching among them. The results clearly vindicated the

van Orden et al. abduction. Each of the time series measured (pupil dilations, heartbeat intervals, keypress durations, and keypress deviations) exhibited $1/f$ noise. As we have seen, this is insufficient to guarantee that the systems in question are interaction dominant. However, there were significant correlations between the physiological measures of $1/f$ noise and between the behavioral measures of $1/f$ noise. This complexity matching is indicative of multifractality in each of these systems, which is to say that it is indicative that they are interaction dominant.

7. Deduction in dynamical cognitive science

According to Peirce, abduction in science ought to be followed by deduction and, eventually, induction. The ongoing deductive phase in complex-systems dynamical cognitive science takes the following form:

1. A consequence of the interaction dominance of the human cognitive system is that we should see complexity matching and multifractality in cognitive phenomena.
2. Test this consequence by looking for complexity matching and multifractality in human cognition.
3. Observing complexity matching and multifractality in human cognition is evidence that human cognitive systems are interaction dominant.

The hypothesis that human cognition is interaction dominant has been tested repeatedly over the last decade and there are literally hundreds of published findings demonstrating complexity matching and multifractality in aspects of human cognition. A small, idiosyncratic selection of these includes individual video game play (Dotov, Nie, & Chemero, 2010; Nie, Dotov, & Chemero, 2011), group jazz performance (Walton, Richardson, Langland-Hassan, & Chemero, 2015, 2018), collaborative video game play (Nalepka, Kallen, Chemero, Saltzman, & Richardson, 2017, 2019, 2021), rhythm production (Coey, Kallen, Chemero, & Richardson, 2018), jewelry making (Baber, Chemero, & Hall, 2019), and perception via sensory substitution devices (Favela, Amon, Lobo, & Chemero, 2021). More generally, we see that multifractality and complexity matching are ubiquitous in physiology, neural activity at every scale, human and animal movement, cognitive tasks, human-tool systems, and human social systems suggesting that each scale is interaction dominant.

This goes beyond the rejection of modularity that we saw in Fodor, Haugeland, and Turvey to a rethinking of the nature and extent of cognitive systems. Recall from above that interaction dominance is indicative of well-functioning physiological systems. This suggests that processes at each scale are as densely interconnected as the components of a human heart. Because these examples of interaction-dominant systems (and many others; e.g., Paxton & Dale, 2013; Abney, Paxton, Dale, & Kello, 2014, 2021) also span human-tool systems and groups of humans, there is good reason to think that extended and multiperson cognitive systems are as real and robust as individual human cognizers. Yet, the cognitive sciences focus almost exclusively on the smallest of these interaction-dominant systems (neurons, brain areas, whole brains), neglecting the larger scale, beyond-the-individual ones (Raja, 2018).

From a complex-systems dynamical point-of-view, there is nothing special about biological or psychological individuals.

8. Coda and exercises for the reader

With so many successful deductions providing evidence for the interaction dominance of human cognitive systems, and even an *experimentum crucis*, we should wonder whether we have arrived at Peirce's induction phase. In induction, one weighs the evidence adduced in favor of an explanation of an observed phenomenon with an eye toward determining whether that evidence is sufficient to make the explanation part of the accepted-but-still-revisable scientific canon. What the criteria for this are is hardly clear. I leave the development of such criteria and the determination of whether the interaction dominance of human cognitive systems is the correct explanation for the ubiquitous multifractality and complexity matching in human behavior as exercises for the reader.

Suppose for the sake of argument that the hypothesis of the interaction dominance of human cognition is in Peirce's induction phase. This would imply that it should appear in textbooks, at the very least, as should discussion of extended human-tool and multiperson cognitive systems. And it now does, thanks to Michael Spivey's additions to the third edition of the textbook *Cognitive Science: An introduction to the study of mind* (Friedenberg, Silverman, & Spivey, 2021). What it does not imply is that other hypotheses about the nature of cognition are false or inadequate. It has certainly not been demonstrated that every aspect of cognitive systems is interaction dominant. It is likely possible to tell similar abduction-to-deduction-to-induction stories for other varieties of dynamical explanation, such as those neural network dynamical explanations that do not align straightforwardly with complex systems theory (e.g., Busemeyer & Townsend, 1993; Izquierdo & Beer, 2009). It is also likely possible to tell abduction-to-deduction-to-induction stories about nondynamical explanations in the cognitive science. I leave telling these stories as another exercise for the reader. What this suggests is that, at least in its current state, cognitive science demands a plurality of explanations (Dale, 2008; Favela & Chemero, 2023). Some aspects of cognition are demonstrably interaction dominant; others might best be explained in terms of neural network dynamics; still others might be best explained in terms of computation. For the time being, despite the progress in complex systems explanations described above, dynamical cognitive science remains a big tent, as does cognitive science as a whole.

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